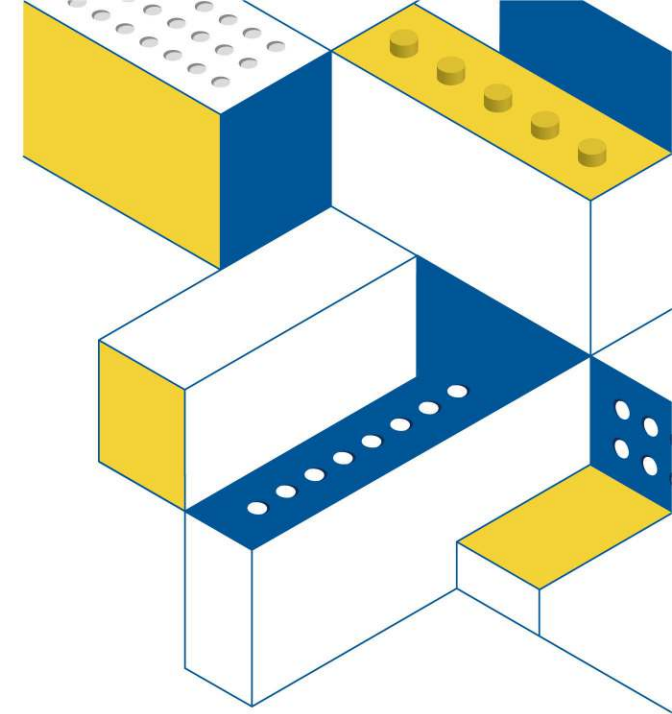


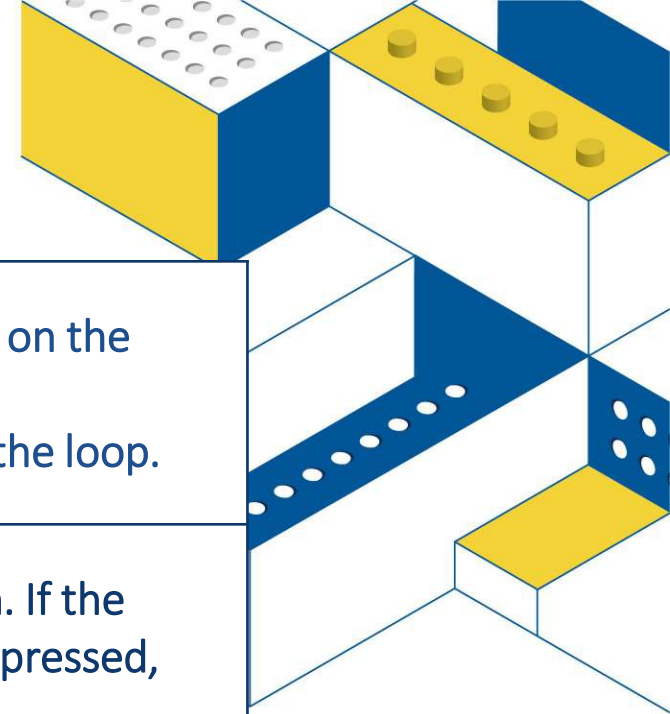
How to Use MATRIX Joystick 2

Learning objectives

- Learn about MATRIX Joystick 2 receiver wiring.
- Understand MATRIX Joystick 2 blocks and value of sticks.
- Understanding how to reset the MJ2.
- How to create variables.
- Understanding mapping block.
- Programming a Remote Control Car.

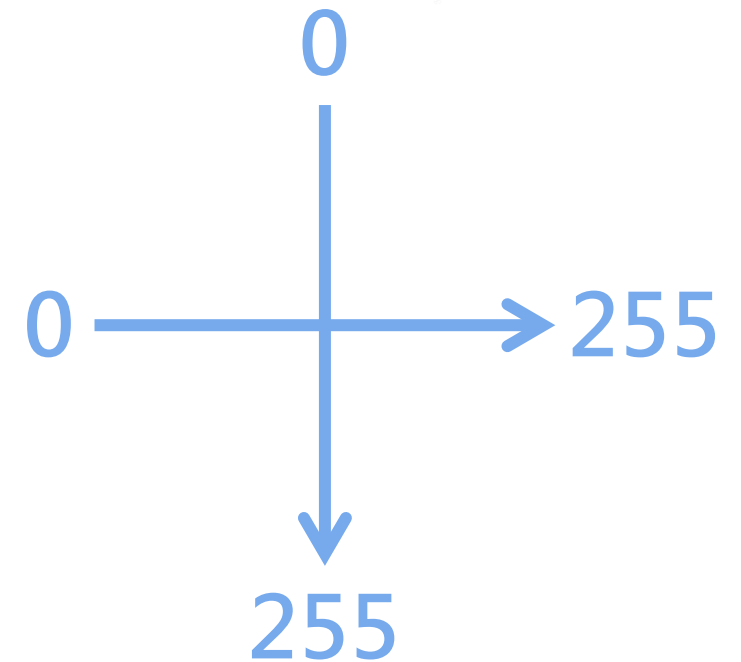
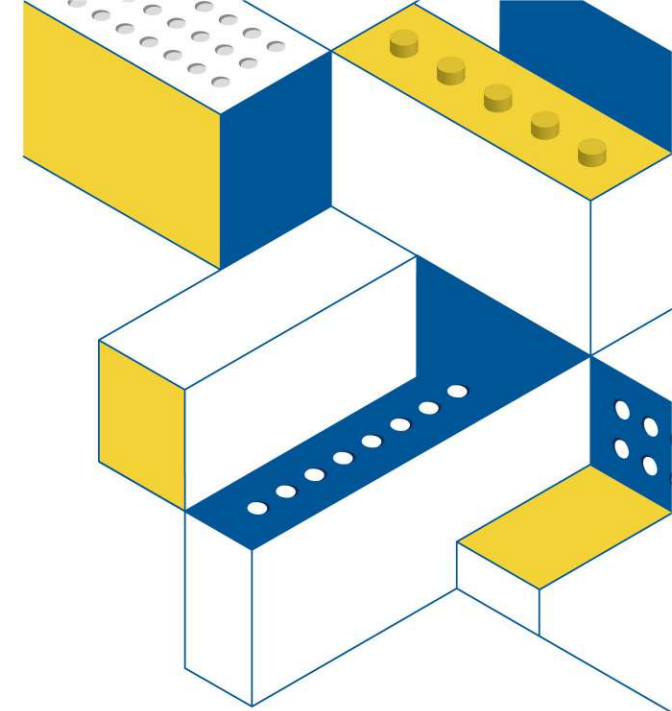


Blocks

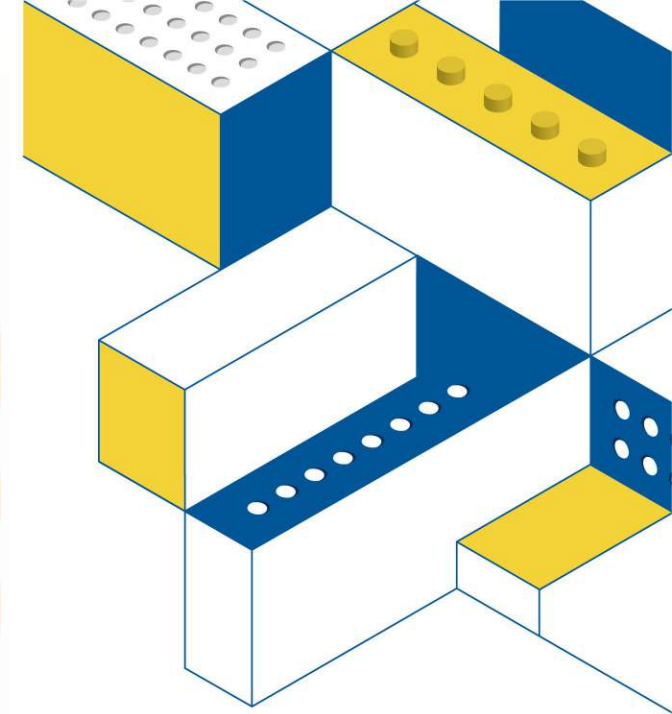
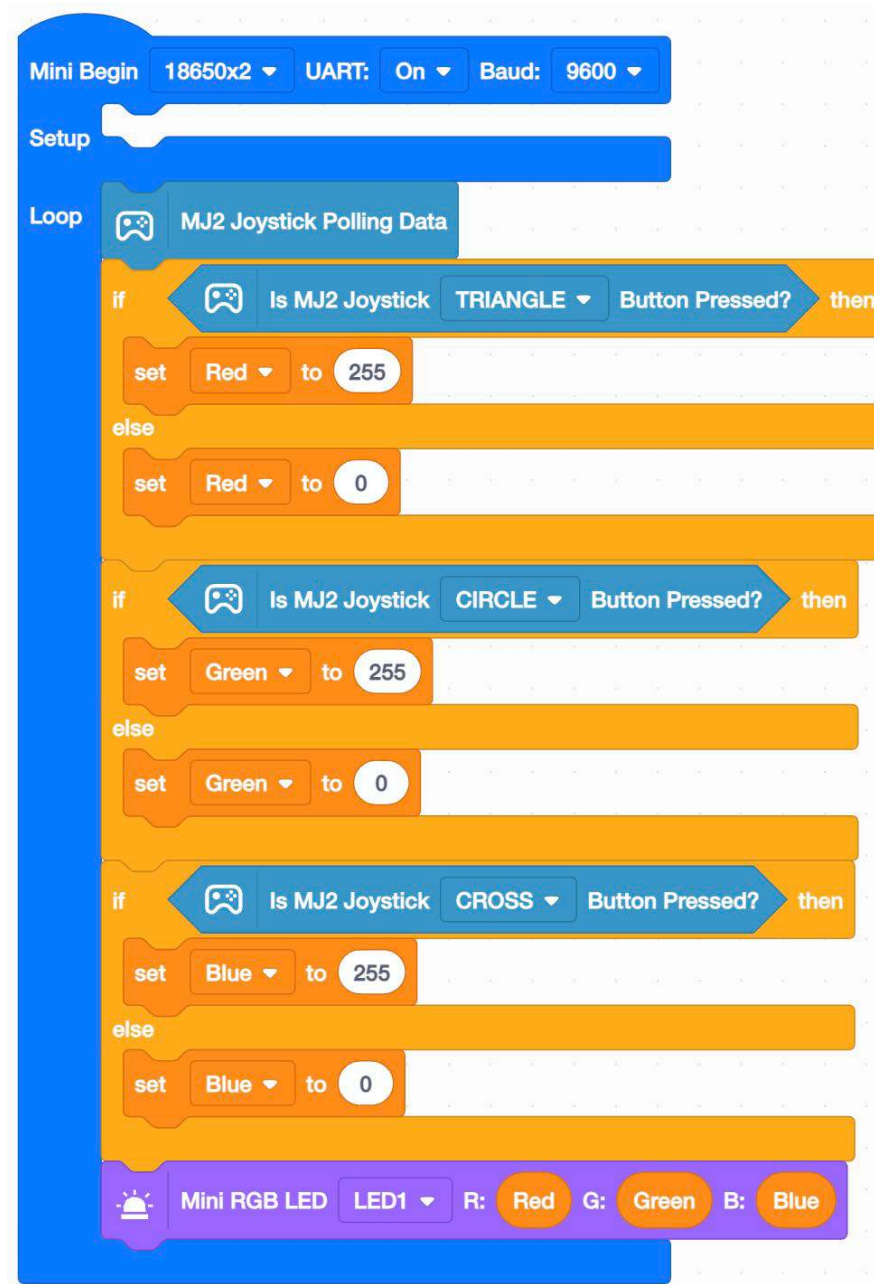


 A blue block with a game controller icon on the left and the text "MJ2 Joystick Polling Data" in white.	Update the status of all buttons and joysticks on the controller. You should put the block in the beginning of the loop.
 A blue arrow-shaped block with a game controller icon on the left, followed by the text "Is MJ2 Joystick", a dropdown menu showing "L1", and the text "Button Pressed?" on the right.	The block represents the status of the button. If the button you choose in the dropdown menu is pressed, the block returns true, vice versa.
 A blue rounded block with a game controller icon on the left, followed by the text "MJ2 Joystick", a dropdown menu showing "Left", another dropdown menu showing "X", and the text "Stick Value" on the right.	The block represents the position of two joysticks in two directions.

Value of two Joysticks



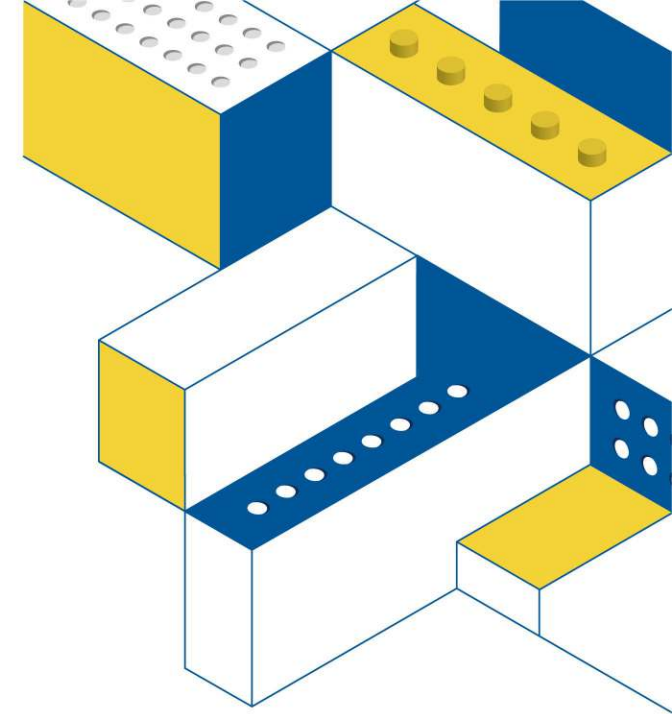
Button Testing



Switch on MJ2

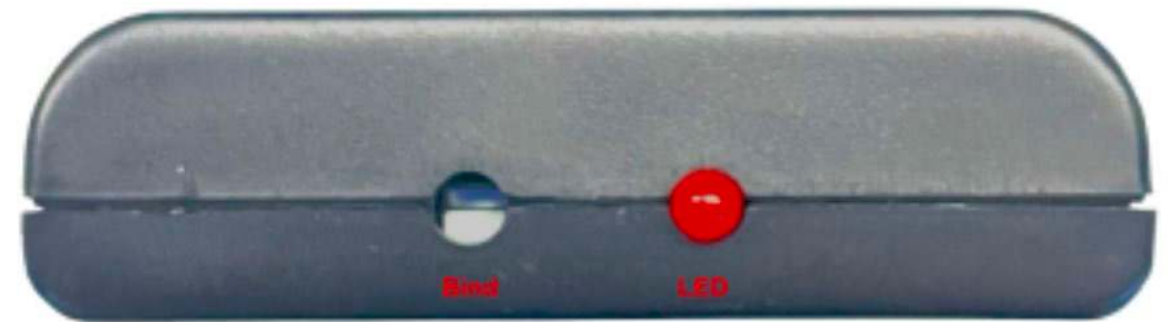
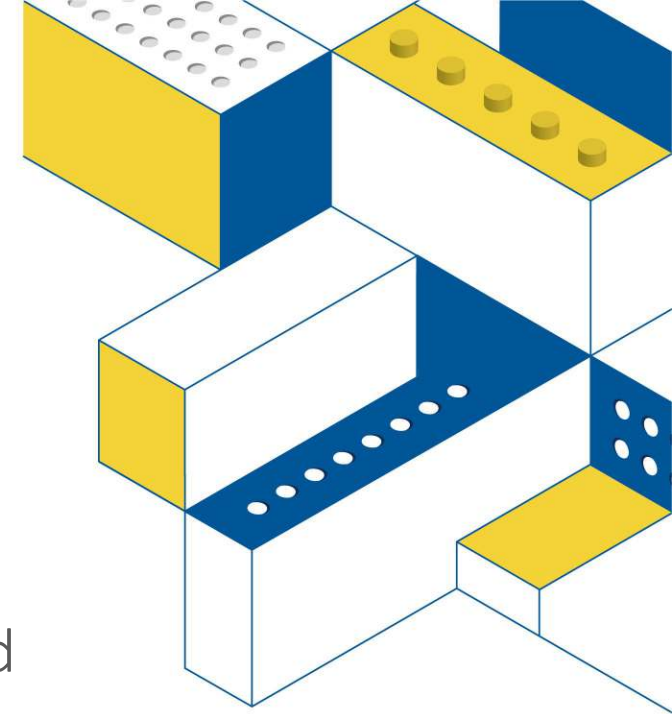
Step1. Install the Battery.

Step2. Press START button until the Red light on.

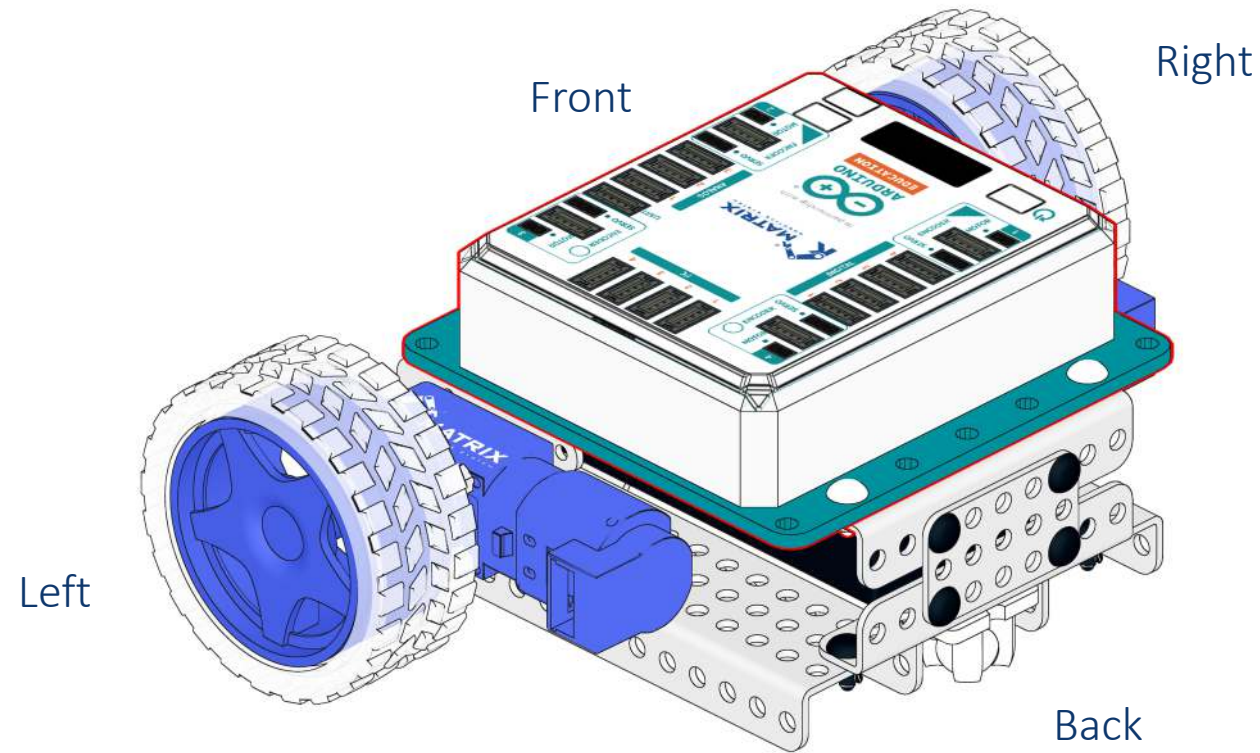


Notice

- Connect the wires of the receiver correctly before turning on the power of the Mini R4 Controller.
- MATRIX Joystick 2 has been paired and set up.
- Press the Bind button of the receiver for more than 2 seconds, and press L3 (left joystick) for 3 seconds to enter the pairing.

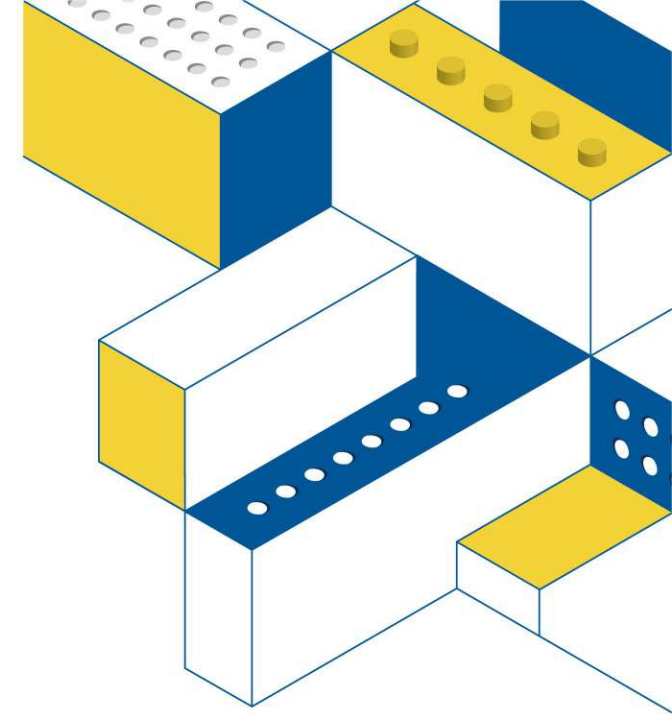
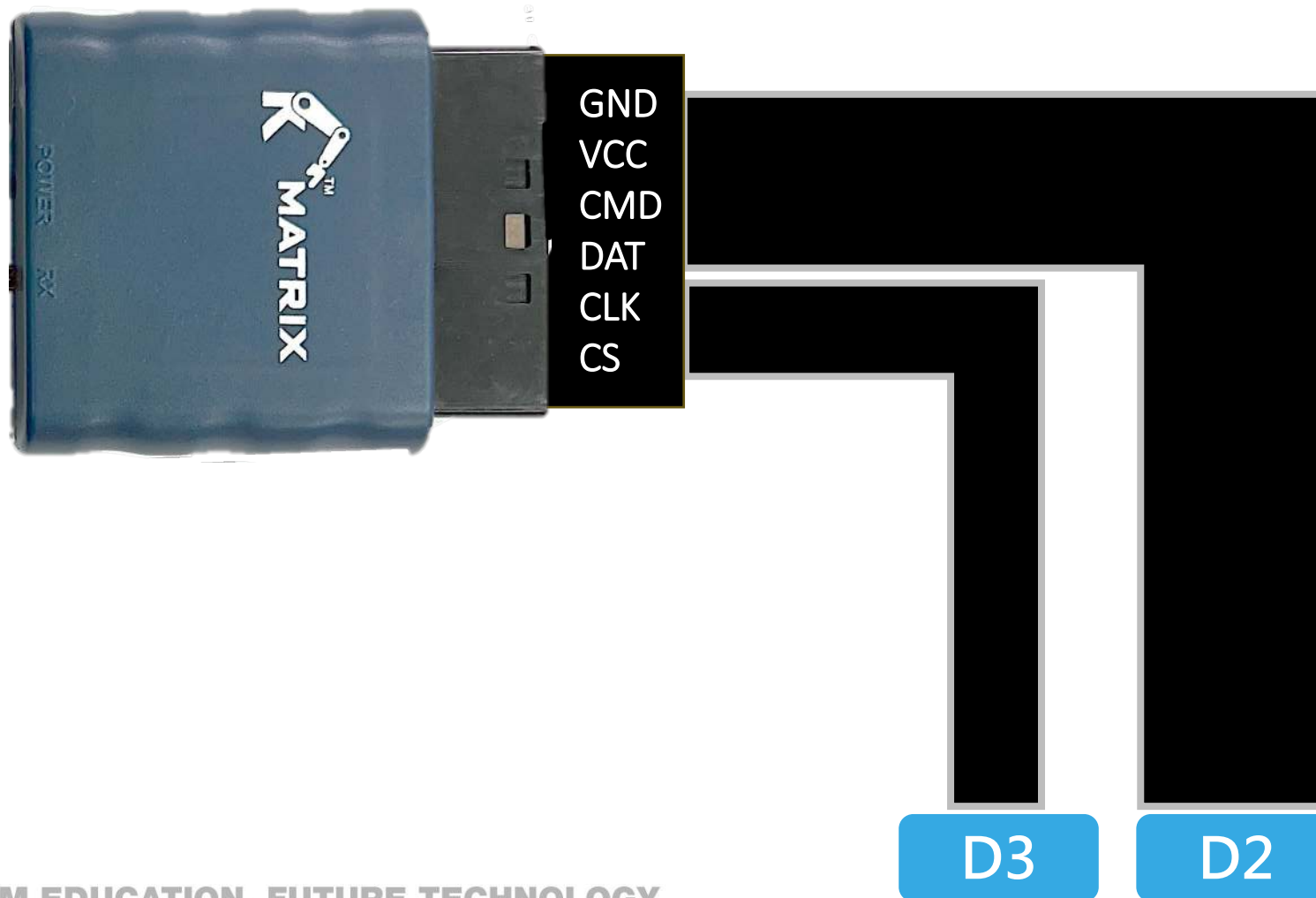


Assembly Basic Car



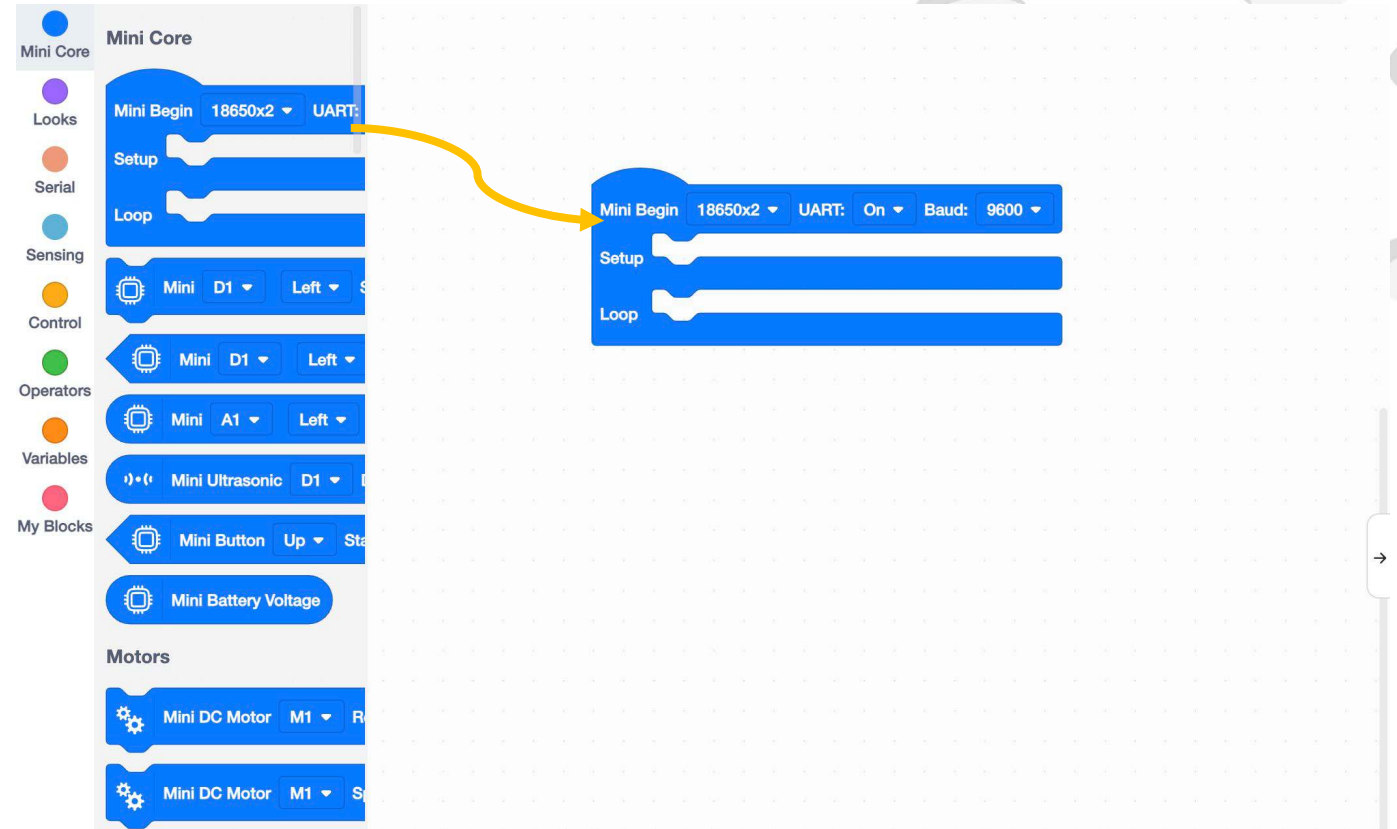
Left motor → Motor 1
Right motor → Motor 2
Black line near green dot "●".

Wiring the receiver and Mini Controller



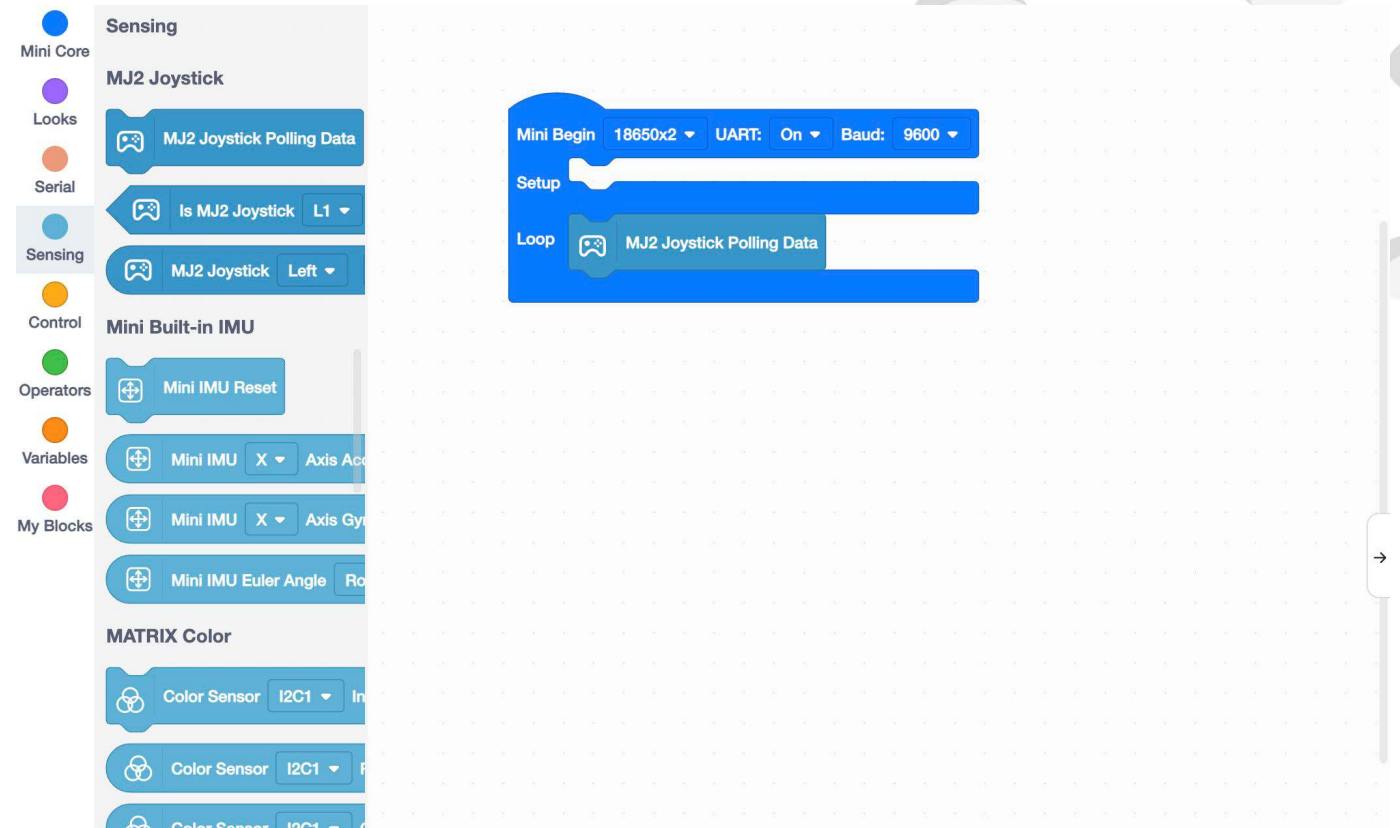
Using MJ2 remote control MATRIX car

1. Drag and drop a Mini Begin block from the Mini Core category into the editing area.



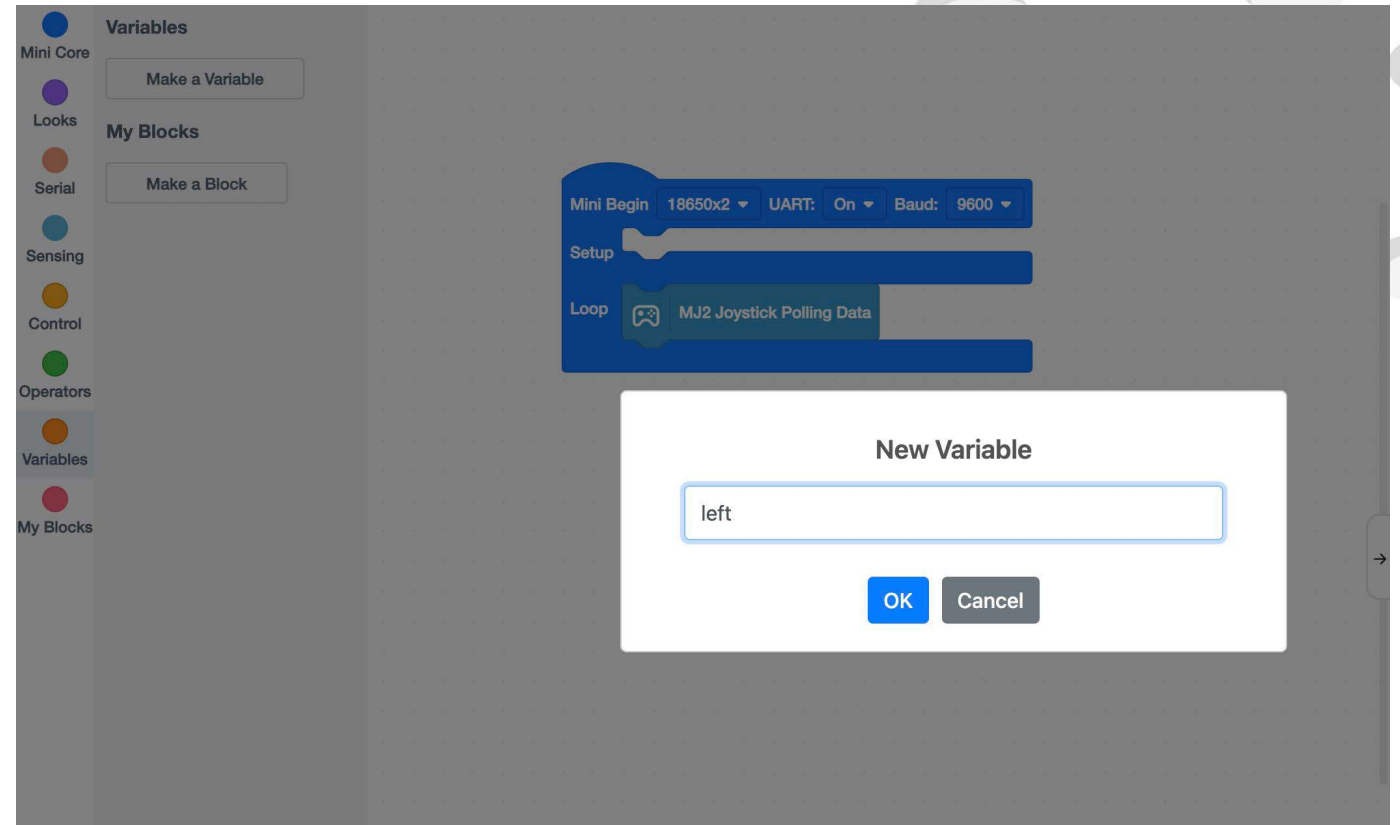
Using MJ2 remote control MATRIX car

2. Drag a MJ2 Joystick Polling Data block from Sensing category into the Loop.



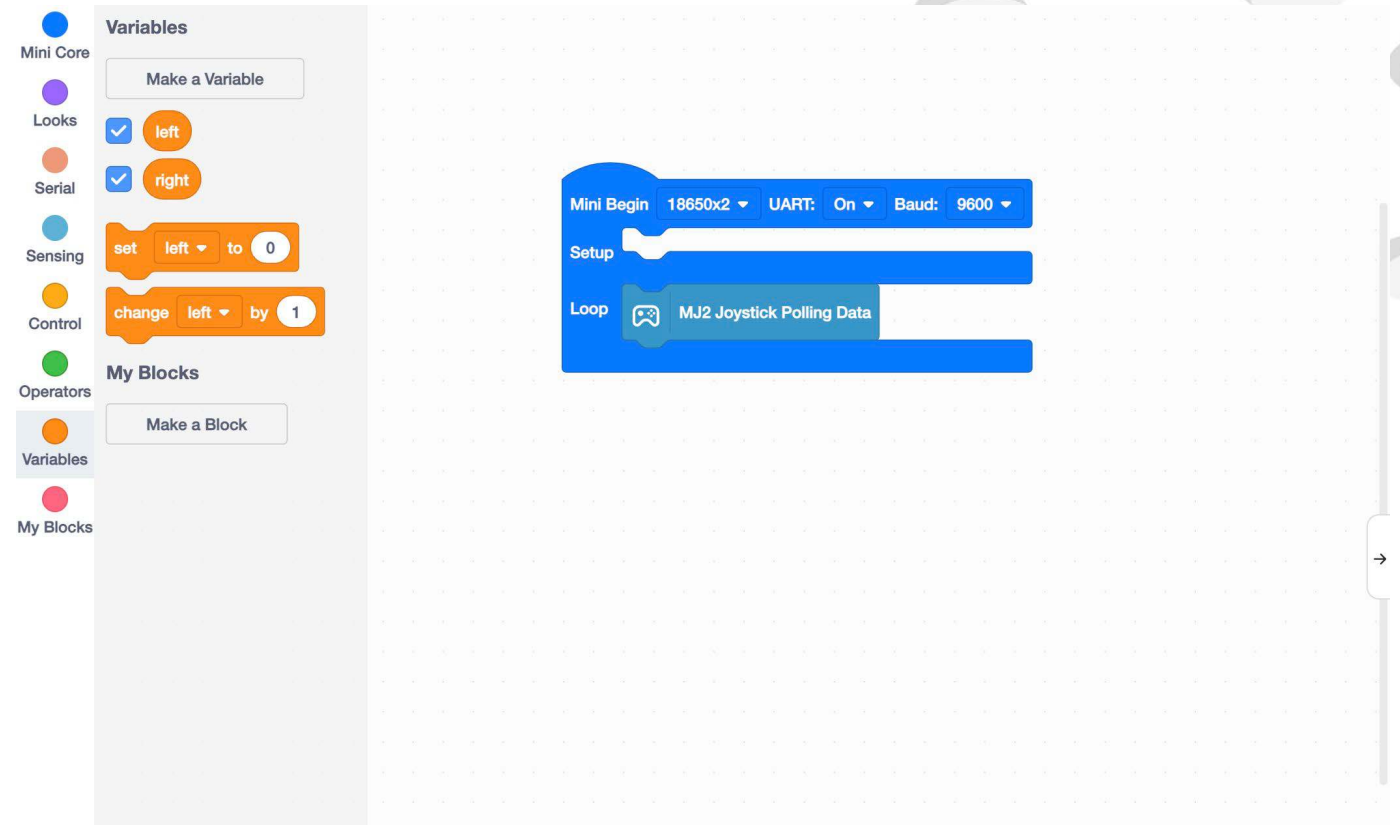
Using MJ2 remote control MATRIX car

3. Create a variable "left" from the category of variables.



Using MJ2 remote control MATRIX car

4. Create another variable "right" from the category of variables.



Using MJ2 remote control MATRIX car

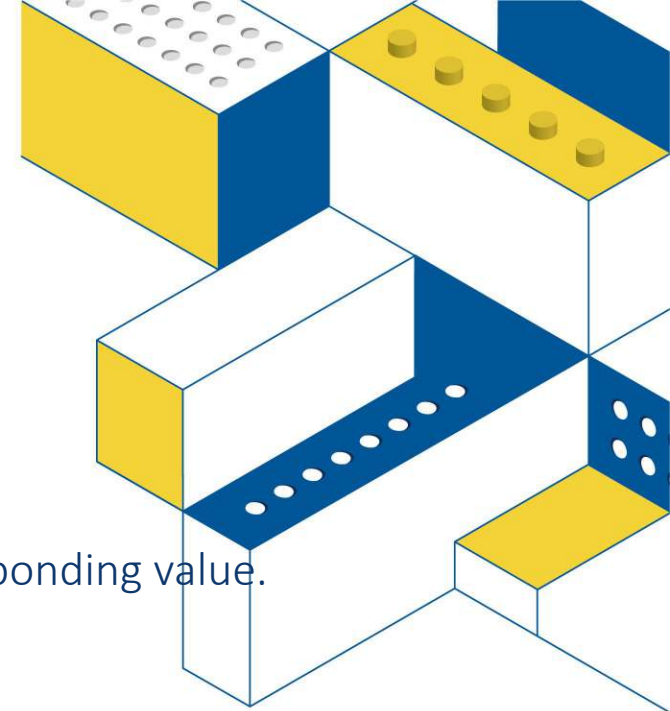
We need to use the Y-axis of the left stick to control the forward and backward movement of the left motor.

When the stick is pushed forward to the end (0), it should correspond to the fastest forward speed (100) of the motor, and when it is pulled back to the end (255), it should correspond to the fastest reverse speed (-100) of the motor.



	MJ2 Value	Motor Speed
Move forward	0	100
Move backwards	255	-100

Map



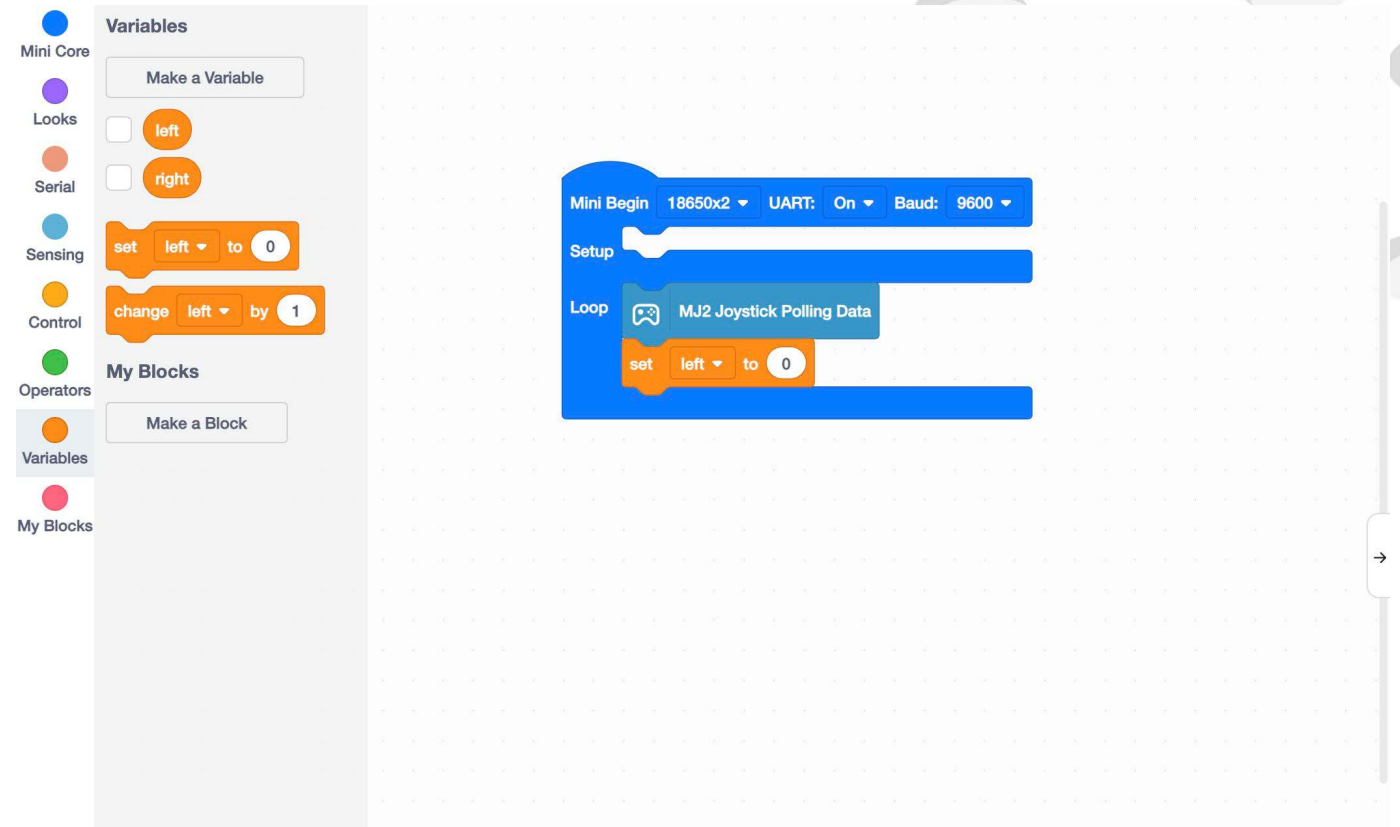
Map → Convert a value to a corresponding value.

The value of the joystick ranges from 0 to 255,
which is converted to 100 to -100 of the motor's power in order to control it.



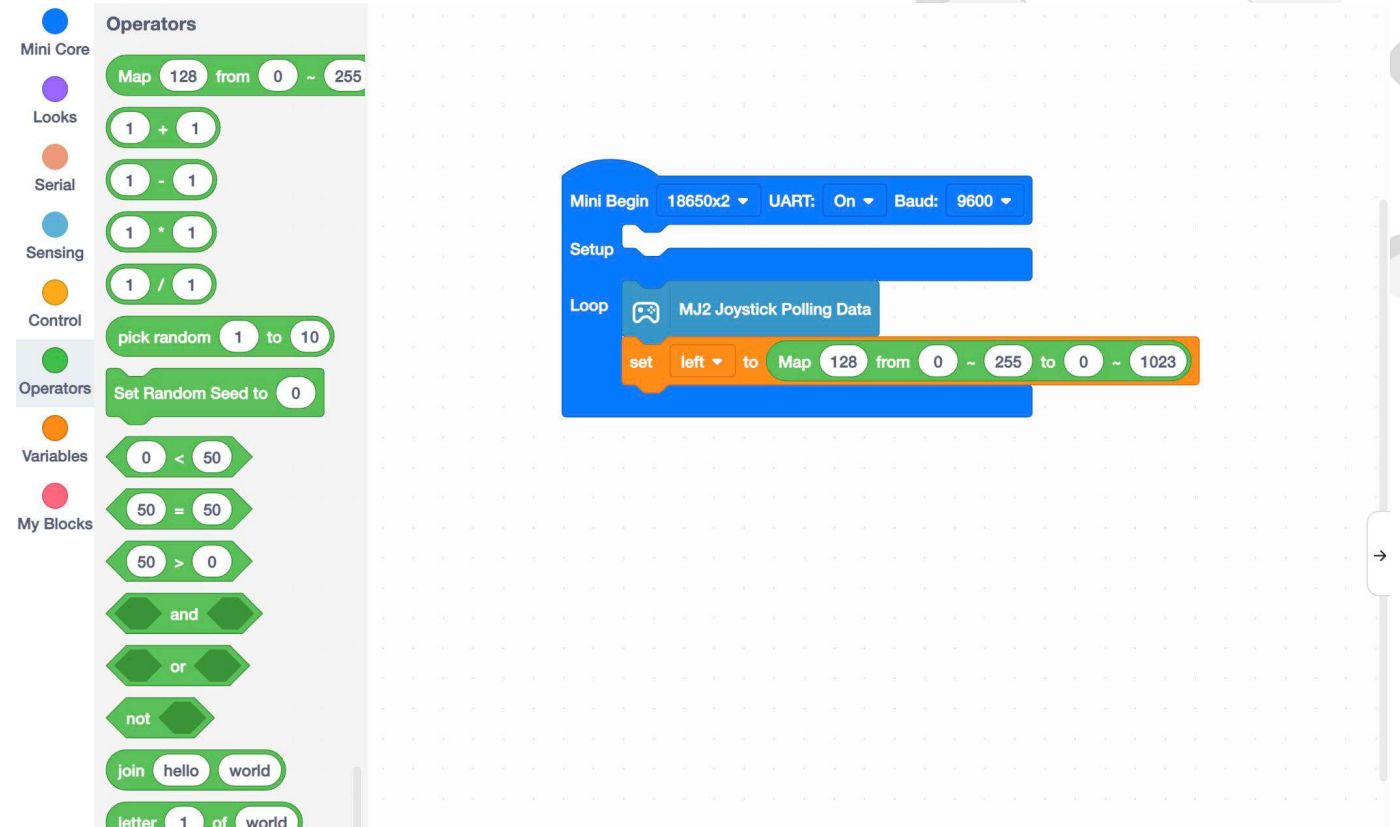
Using MJ2 remote control MATRIX car

5. Drag a "set left to" block from the variables category to the edit area.



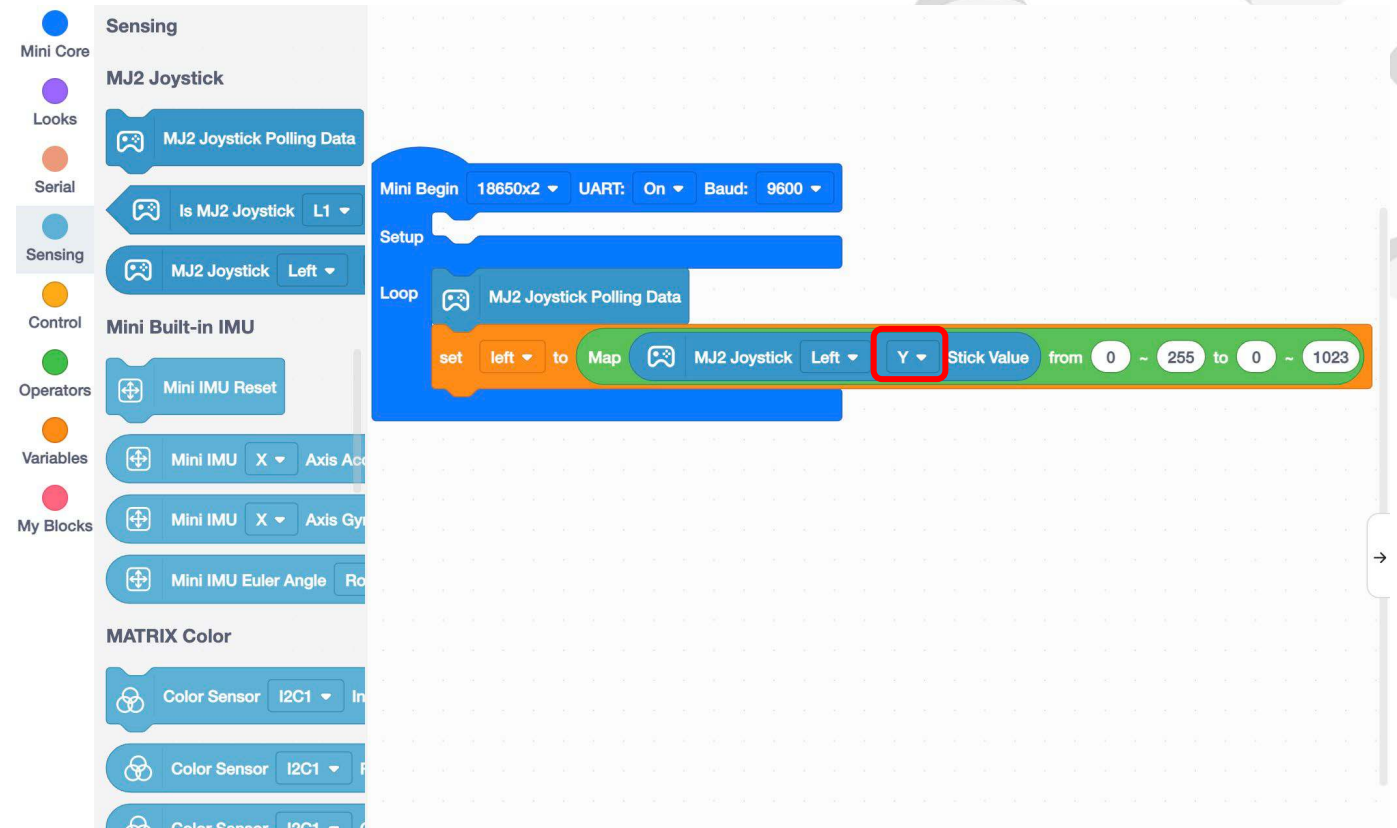
Using MJ2 remote control MATRIX car

6. Drag a Map block from the Operators category to the editing area.



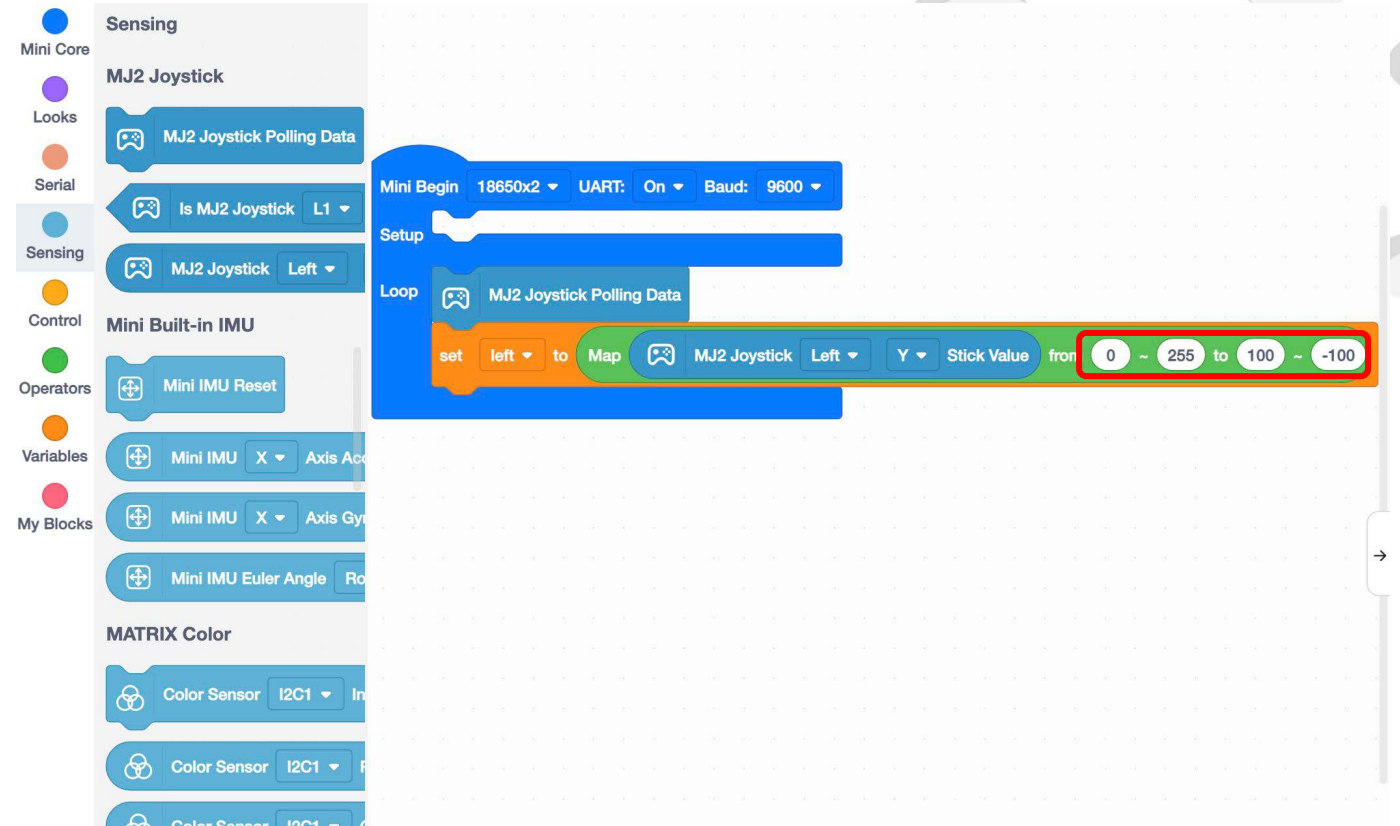
Using MJ2 remote control MATRIX car

7. Drag a block of MJ2 stick value from the Sensing category to the first input of the map and select the Y-axis.



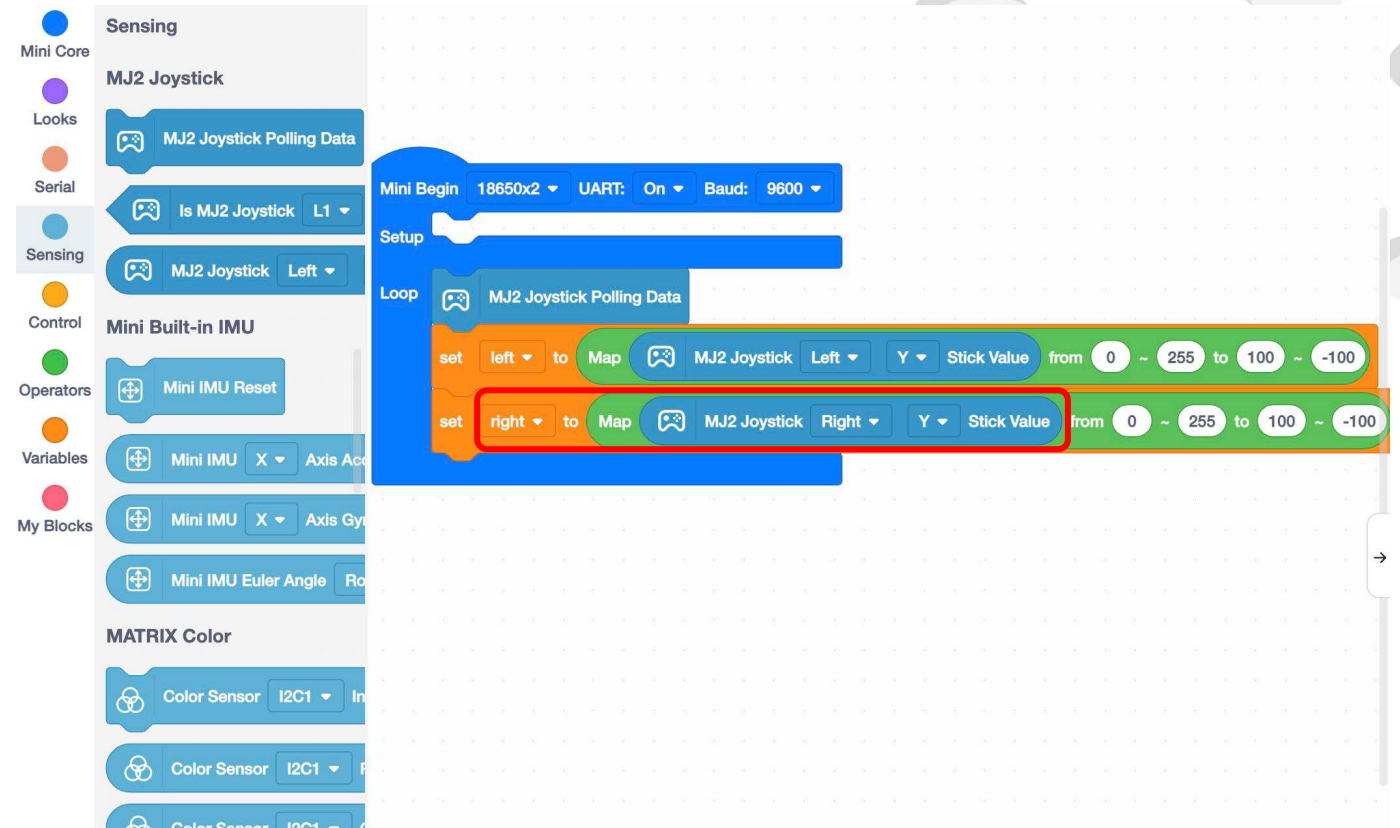
Using MJ2 remote control MATRIX car

8. Enter the values of the input cells in the mapping block in order: 0 → 255 → 100 → -100



Using MJ2 remote control MATRIX car

9. The variable "right" follows the procedure of left.



Using MJ2 remote control MATRIX car

10. Drag the 2 Mini DC Motor Reverse blocks to Setup and set M2 reverse.



Using MJ2 remote control MATRIX car

11. Use the variable left for the speed of DC motor M1; use the variable right for the speed of DC motor M2.

The image shows a Scratch script for controlling a MATRIX car. The script is organized into three sections: Mini Begin, Setup, and a Loop.

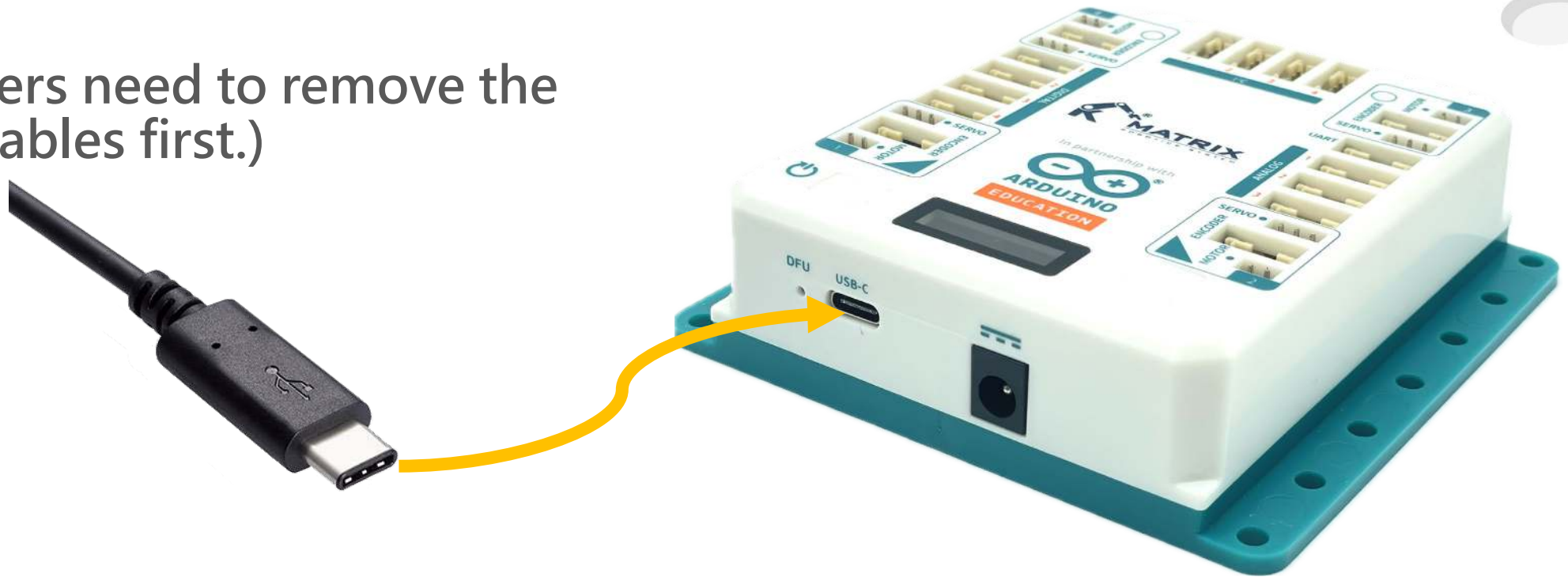
- Mini Begin:** Configures the Mini Core with 18650x2 batteries, UART On, and a Baud rate of 9600.
- Setup:** Initializes two Mini DC Motors. Motor M1 is set to Reverse No, and Motor M2 is set to Reverse Yes.
- Loop:** A continuous loop that polls MJ2 joystick data. It contains two 'set' blocks that map joystick stick values to variables 'left' and 'right'. The 'left' variable is mapped from the Left Y-axis (0 to 255 to 100 to -100), and the 'right' variable is mapped from the Right Y-axis (0 to 255 to 100 to -100). Below these, two 'Mini DC Motor' blocks are shown, each with a 'Speed' dropdown set to the corresponding variable ('left' for M1 and 'right' for M2). These two motor control blocks are highlighted with a red rectangle.

The left sidebar shows the 'Variables' category selected, with 'left' and 'right' variables already created. The 'My Blocks' section also shows a 'Make a Block' button.

Using MJ2 remote control MATRIX car

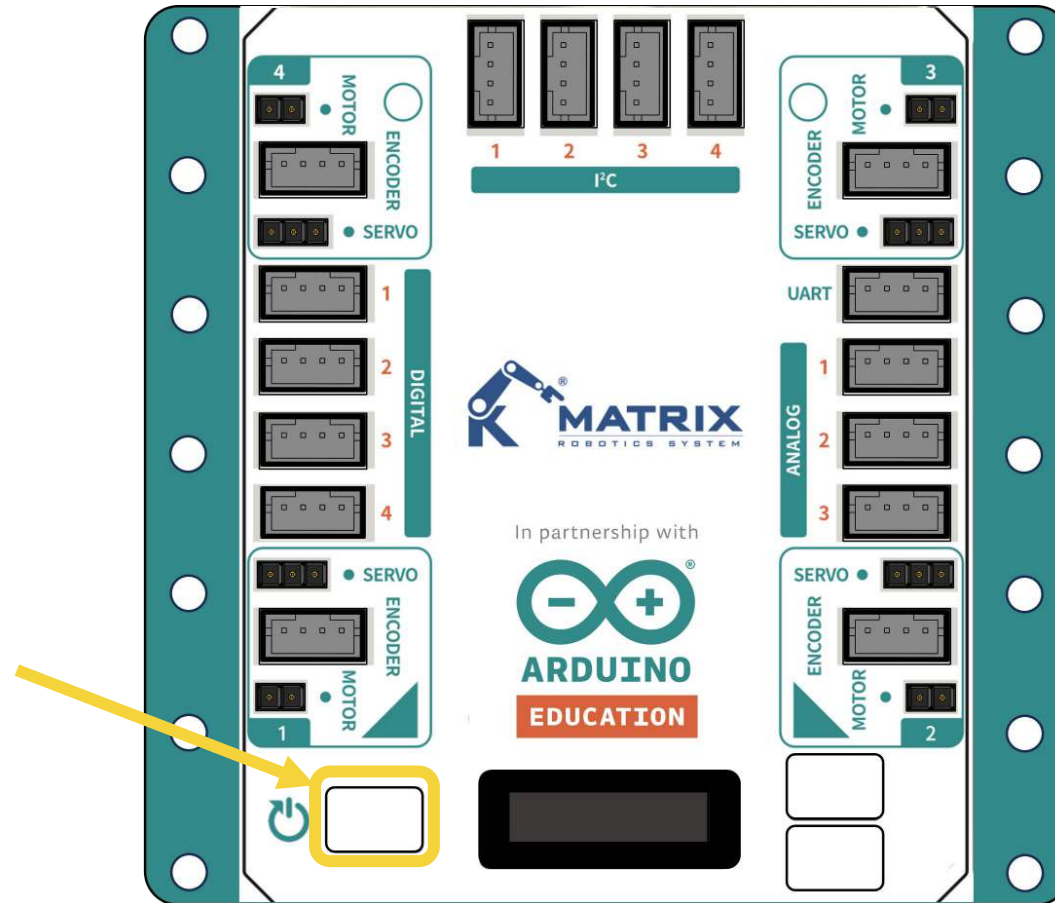
12. Connect the controller to the computer using a USB cable.

(Mac users need to remove the power cables first.)



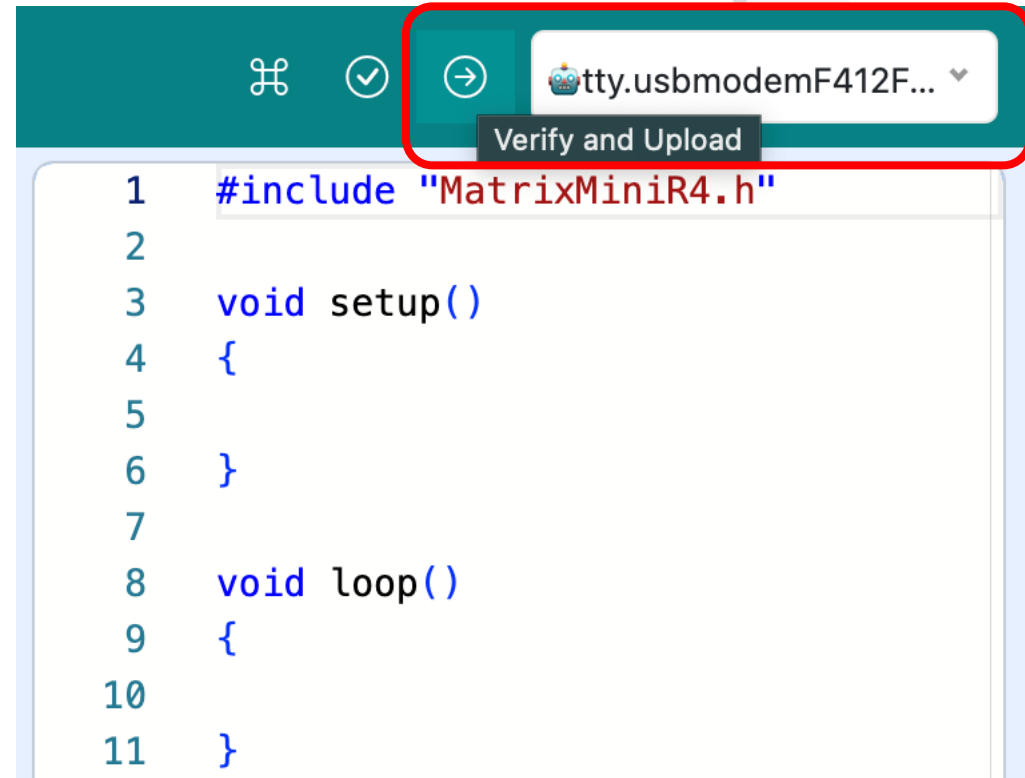
Using MJ2 remote control MATRIX car

13. Press and hold the Reset Button to switch on the Mini R4.



Using MJ2 remote control MATRIX car

14. Select the COM port and upload the program.



Using MJ2 remote control MATRIX car

15. Connect the battery box and press the reset button to run the program.

(Make sure the battery box has switched on)





**Congratulations on
completing all units!**

**Have fun designing your own
robots!**